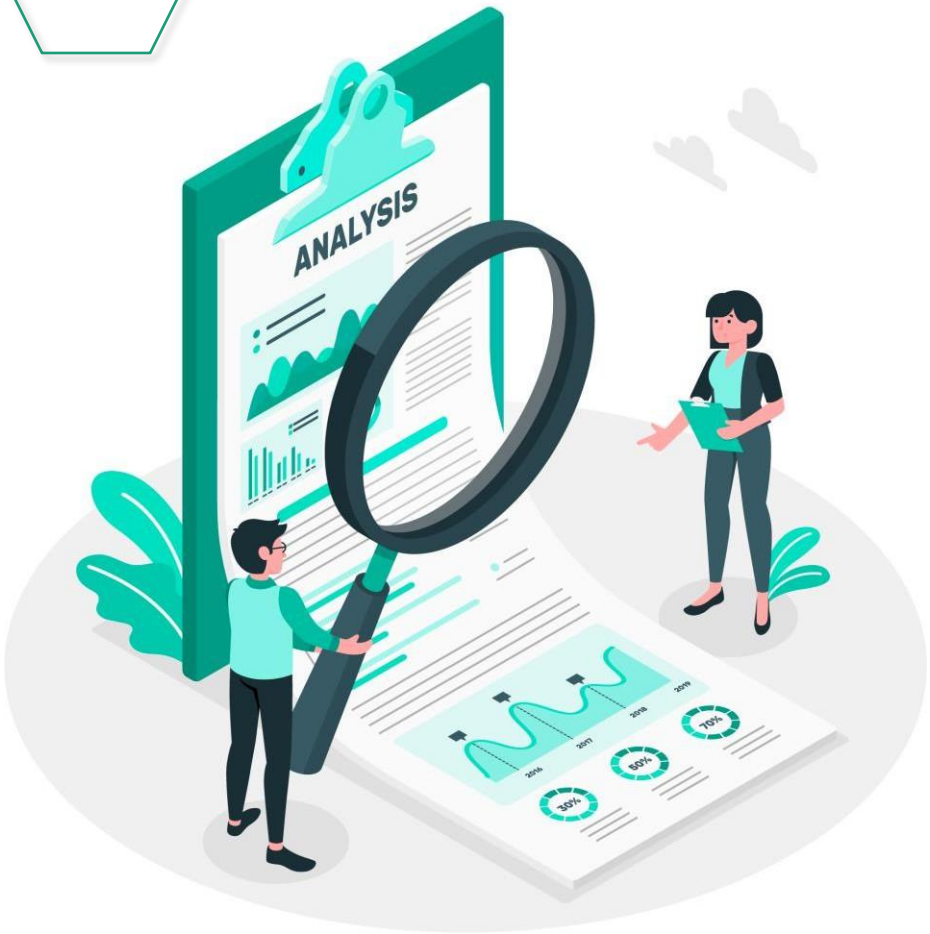
My LinkedIn




Atliq Hardwares

# Consumer Goods Ad-hoc Insights

Presented by **Bhavana T**

SQL Project Challenge

# AGENDA

1. Background/Context
2. Getting familiar with Atliq's Business – Their Markets and Product lines
3. Getting familiar with the input data
4. Ad-hoc requests along with the queried results, visualizations and Insights

1

# BACKGROUND/CONTEXT

## Our Company

Atliq Hardwares (imaginary company) – One of the leading computer hardware producers in India.

## Background

The management noticed that they do not get enough insights to make quick and smart data-informed decisions.

## Problem

There are 10 ad-hoc requests for which the company needs insights.

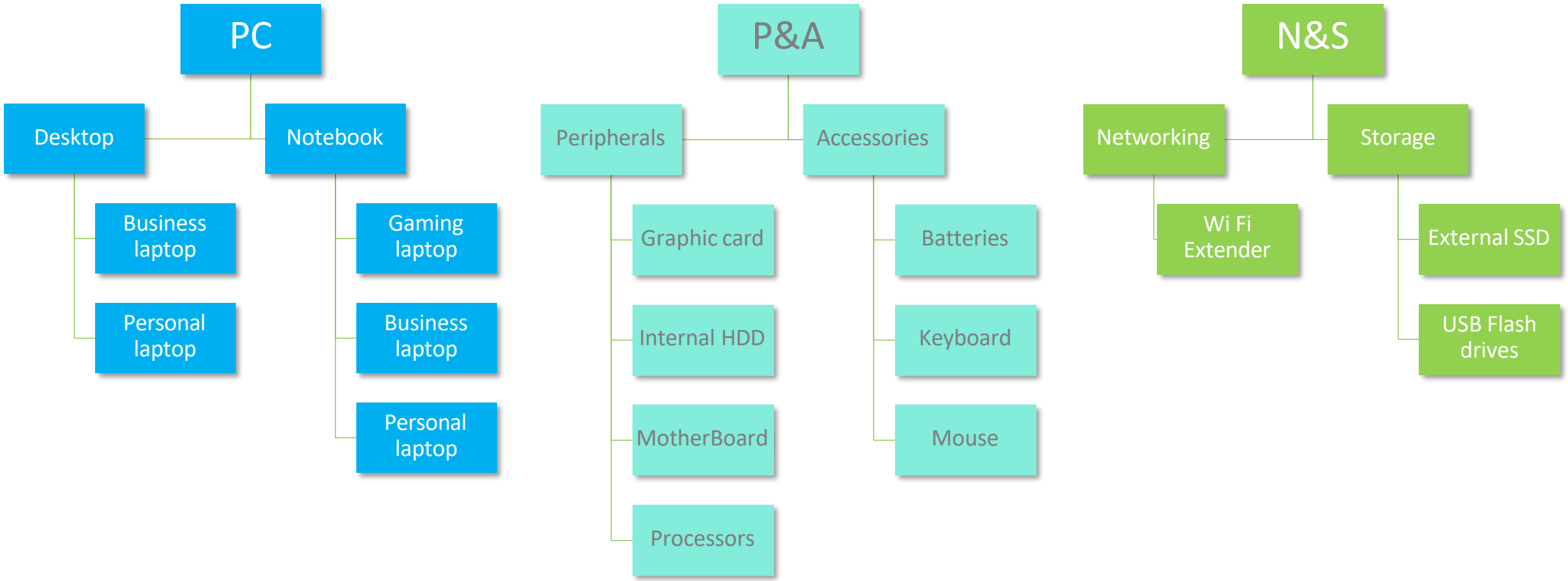
## Approach

Run a SQL query to answer these requests. Convert it into visualizations and present the Insights to the top-level management.

# 2 Getting familiar with Atliq's Business –Their Markets and Product lines

## Atliq's Markets

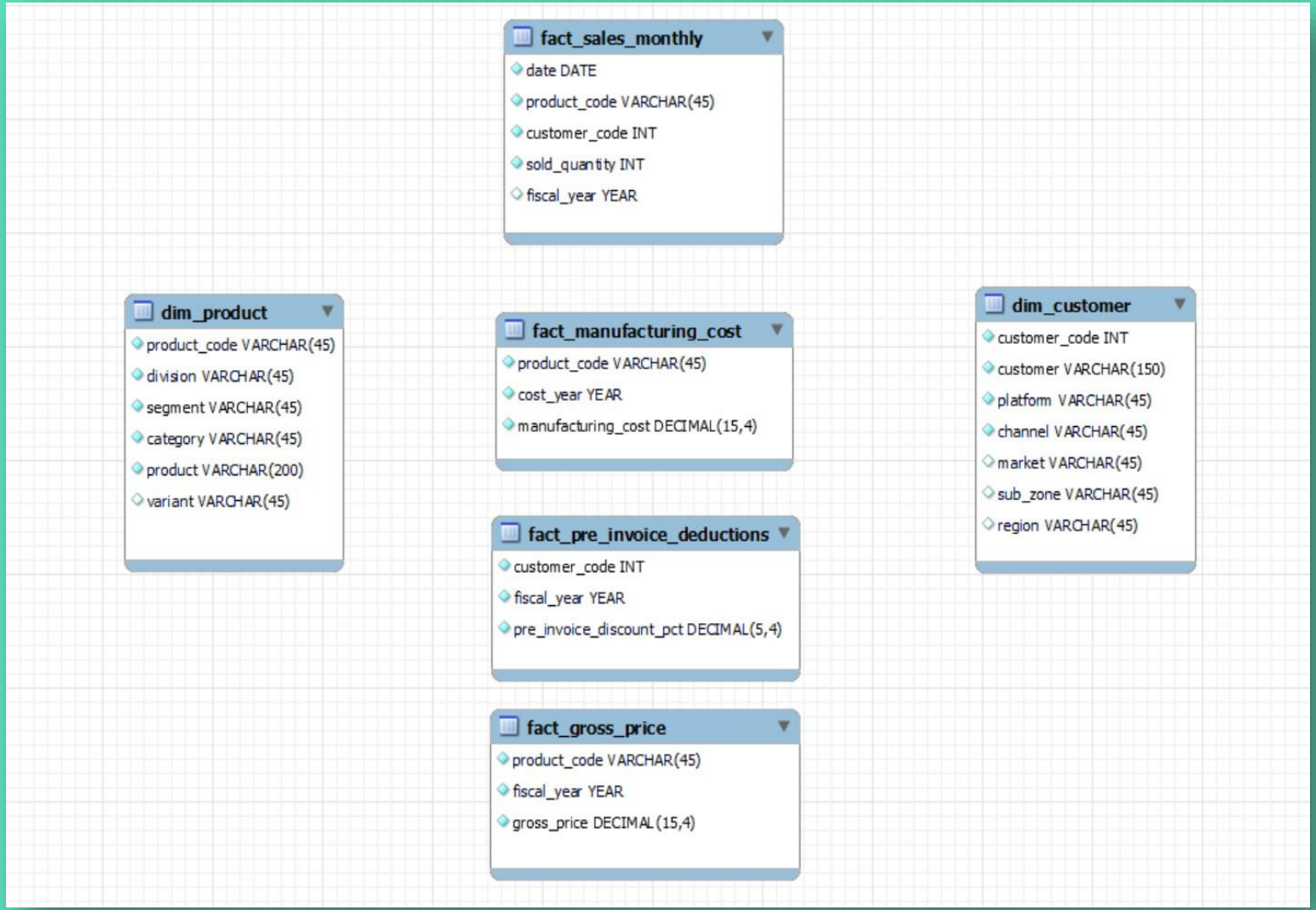
# Atliq’s Product lines




Created by Bhavana T

# 3 Getting familiar with the input data

- Input data consists of sales data for FY 2020 and FY 2021, along with different other dimension tables like customer details, product details, etc.



4

# Ad-hoc requests, queried results, Insights and visualization

# REQUEST 1:

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

# OUTPUT

| market |
|---|
| India |
| Indonesia |
| Japan |
| Philiphines |
| South Korea |
| Australia |
| Newzealand |
| Bangladesh |

# Insights

In the APAC region, our Exclusive store has established its presence in 8 major markets.

# REQUEST 2:

What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields:

unique_products_2020
unique_products_2021
percentage_chg

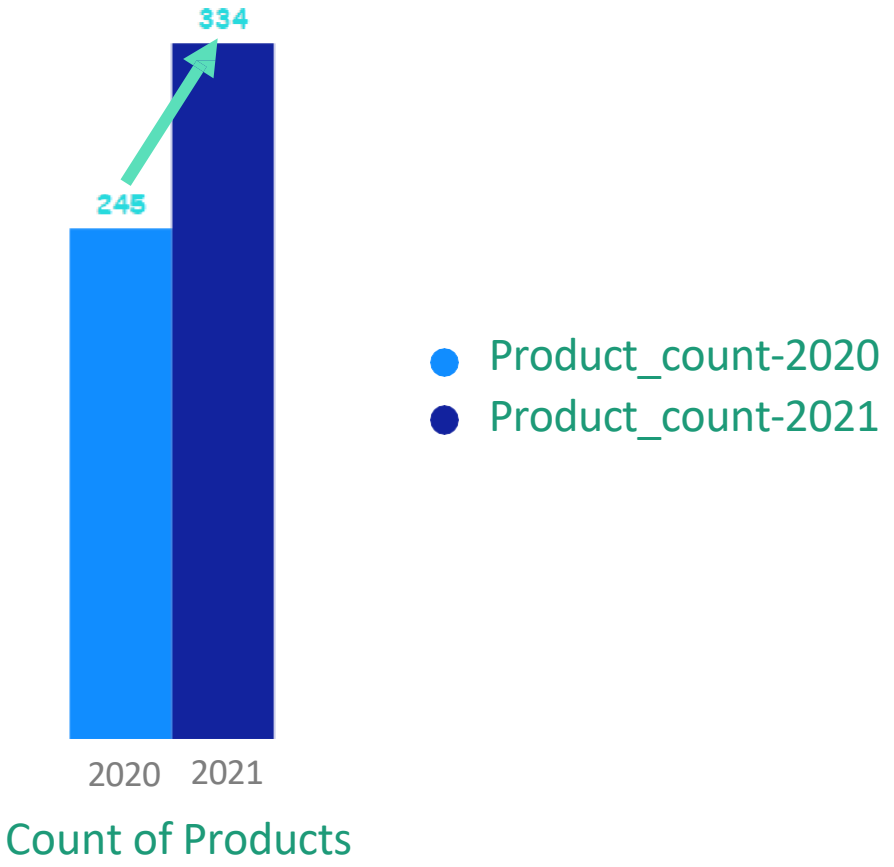
# OUTPUT

| | unique_products_2020 | unique_products_2021 | percentage_change |
|---|---|---|---|
| ▶ | 245 | 334 | 36.33 |

# Insights

It’s a good sign that we are continuously innovating and introducing new products to the market. **In FY 2020, we had a total of 245 products**, but **in FY 2021, our count increased by 36% to 334 products.**

**36.33**
% increase




Count of Products

# REQUEST 3:

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields:

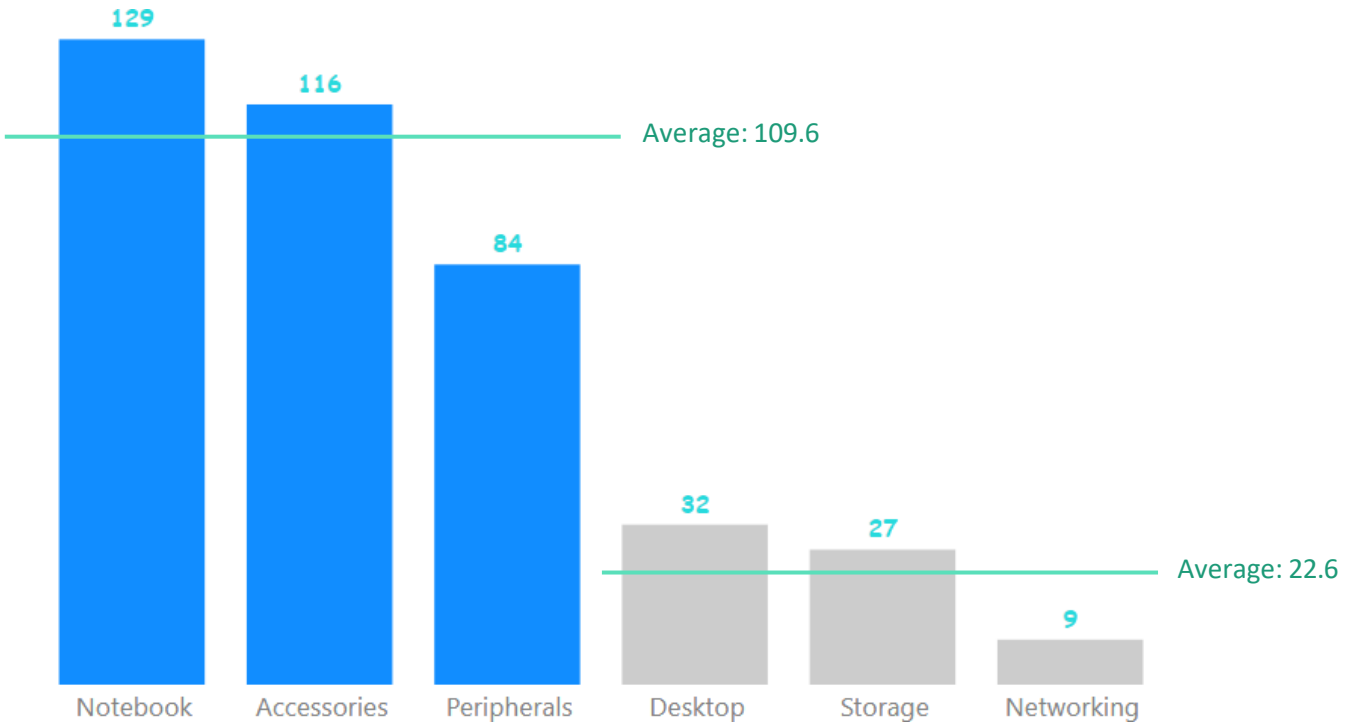
Segment
product_count

# OUTPUT

| segment | product_count |
|---|---|
| Notebook | 129 |
| Accessories | 116 |
| Peripherals | 84 |
| Desktop | 32 |
| Storage | 27 |
| Networking | 9 |

# Insights

We provide a wide range of products under the segments **Notebook, Peripherals, and Accessories, with an average of 110 products in each segment**. However we still need to diversify our production in the **Desktop, Networking, and Storage segments,** where there are just an **average of only 23 products per segment**.




## How many products are there in each segments?

# REQUEST 4:

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields:

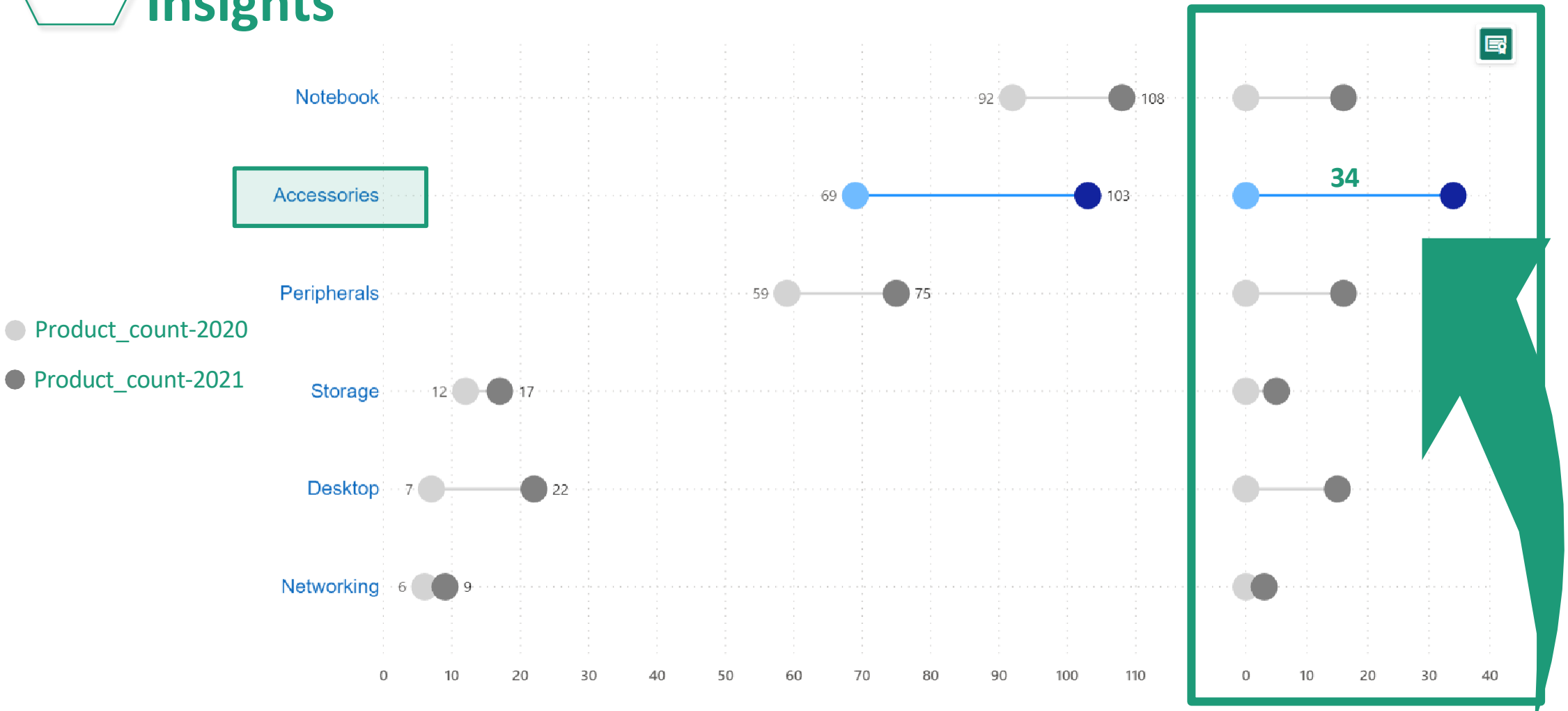
segment
product_count_2020
product_count_2021
difference

# OUTPUT

| segment | product_count_2020 | product_count_2021 | difference |
|---|---|---|---|
| Accessories | 69 | 103 | 34 |
| Notebook | 92 | 108 | 16 |
| Peripherals | 59 | 75 | 16 |
| Desktop | 7 | 22 | 15 |
| Storage | 12 | 17 | 5 |
| Networking | 6 | 9 | 3 |

# Insights

In 2021, we were mainly focusing on diversifying **our accessories segment.**
We introduced **34 new products** to the market in accessories.




Comparison of Product count in 2020 vs 2021 for each of the segments

Which segment had the most increase?

Created by Bhavana t

# REQUEST 5:

Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields:

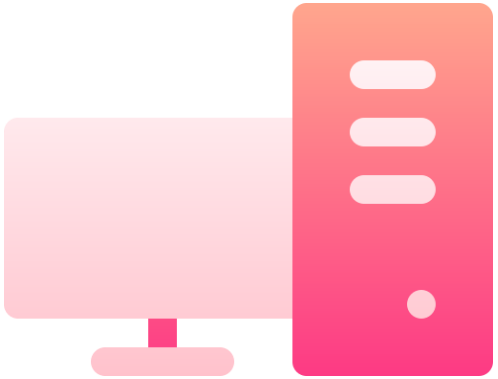
product_code
product
manufacturing_cost

# OUTPUT

| product_code | product | cost_year | manufacturing_cost |
|---|---|---|---|
| A6120110206 | AQ HOME Allin1 Gen 2 (Plus 3) | 2021 | 240.5364 |
| A2118150101 | AQ Master wired x1 Ms (Standard 1) | 2020 | 0.8920 |

# Insights

**Which of our products has the highest manufacturing cost?**



240.54$

AQ HOME Allin1 Gen 2 (Plus 3)

**Personal Desktop**

**Which of our products has the lowest manufacturing cost?**

0.89$

AQ Master wired x1 Ms (Standard 1)

**Mouse**

# REQUEST 6:

Generate a report which contains the top 5 customers who received an average high pre_invoice discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields:

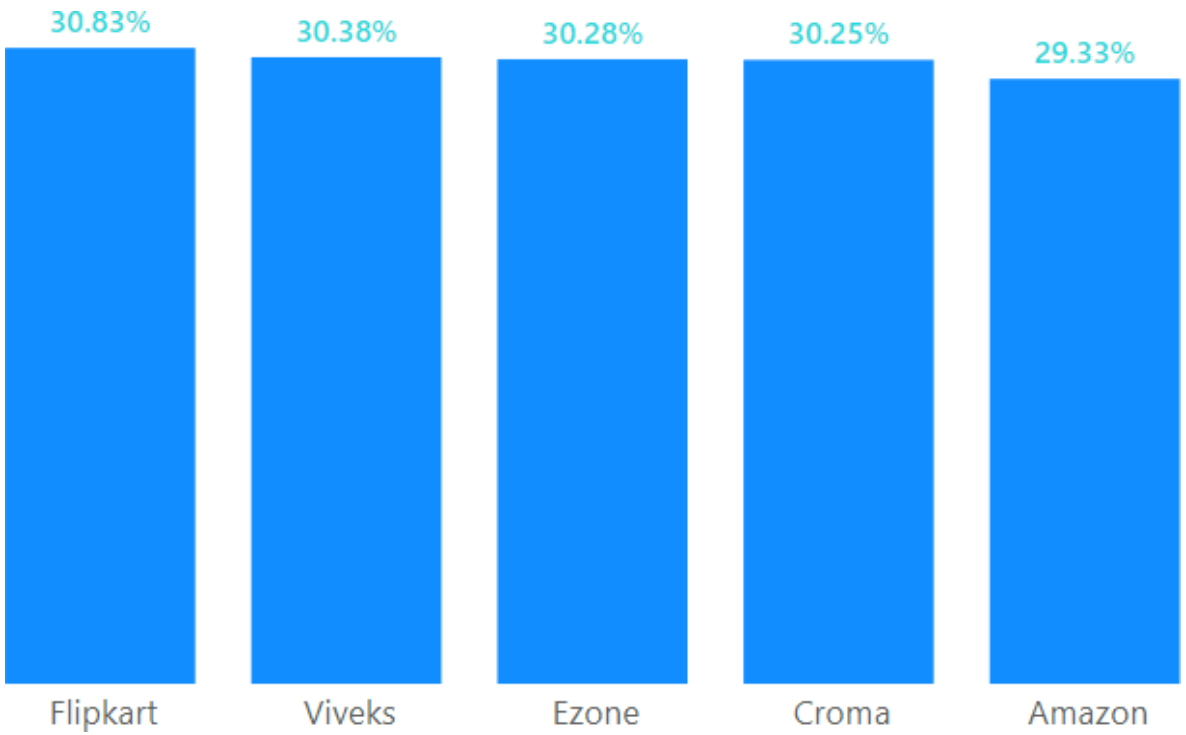
customer_code
customer
average_discount_percentage

# OUTPUT

| customer_code | customer | average_discount_percentage |
|---|---|---|
| 90002009 | Flipkart | 0.3083 |
| 90002006 | Viveks | 0.3038 |
| 90002003 | Ezone | 0.3028 |
| 90002002 | Croma | 0.3025 |
| 90002016 | Amazon | 0.2933 |

# Insights

In 2021, **we offered nearly equal pre-invoice discount percentages** to each of **our top 5 customers,** given that **Flipkart** is the most discounted customer in the Indian market, which equals **30.83%.**




Who are the top 5 customers for whom we offered the most pre-invoice discount percentage in 2021 and in the Indian market?

# REQUEST 7:

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns:

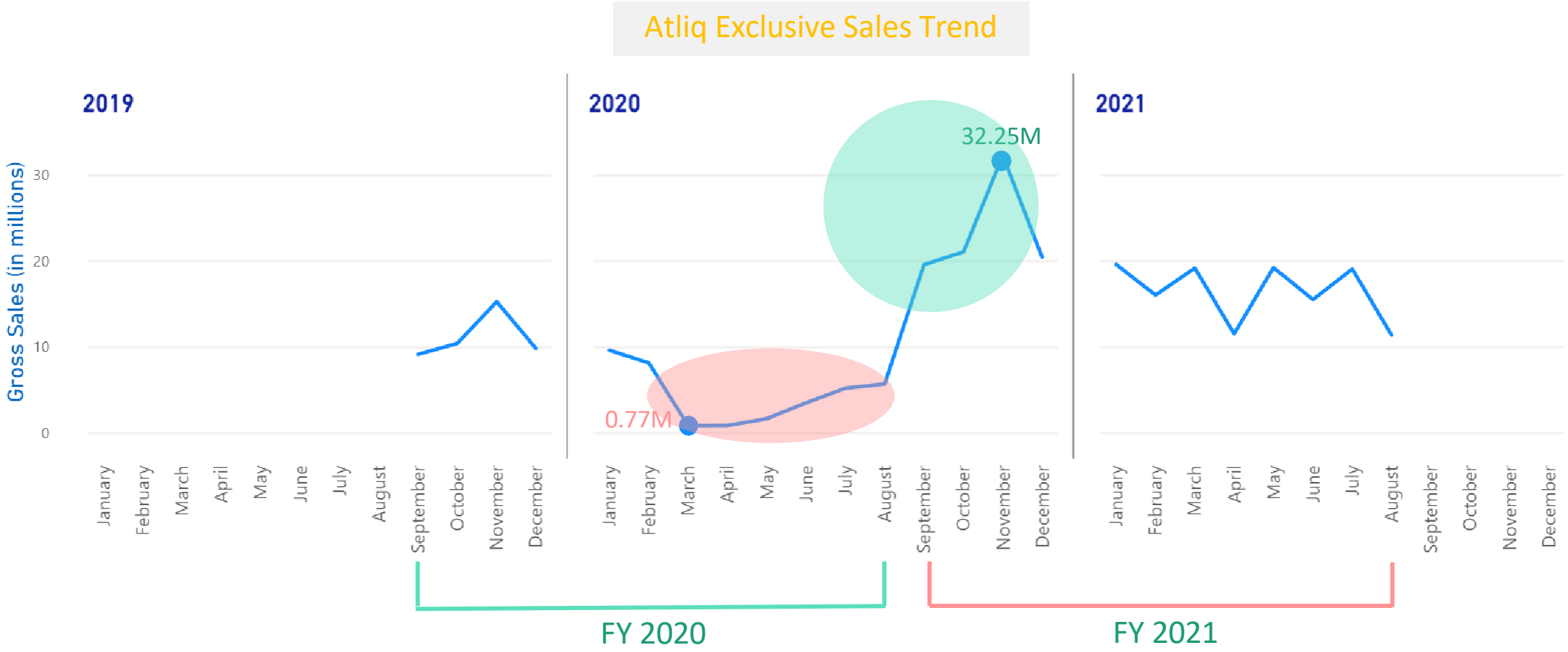
Month
Year
Gross sales Amount

# OUTPUT

| months | year | gross_sales |
|---|---|---|
| September | 2019 | 9.09M |
| October | 2019 | 10.38M |
| November | 2019 | 15.23M |
| December | 2019 | 9.76M |
| January | 2020 | 9.58M |
| February | 2020 | 8.08M |
| March | 2020 | 0.77M |
| April | 2020 | 0.80M |
| May | 2020 | 1.59M |
| June | 2020 | 3.43M |
| July | 2020 | 5.15M |
| August | 2020 | 5.64M |
| September | 2020 | 19.53M |
| October | 2020 | 21.02M |
| November | 2020 | 32.25M |
| December | 2020 | 20.41M |
| January | 2021 | 19.57M |
| February | 2021 | 15.99M |
| March | 2021 | 19.15M |
| April | 2021 | 11.48M |
| May | 2021 | 19.20M |
| June | 2021 | 15.46M |
| July | 2021 | 19.04M |
| August | 2021 | 11.32M |

# Insights

For Atliq Exclusive, **November 2020** marked the **highest sales**, and **March 2020** marked the **lowest gross sales**. It's very evident that the **lower sales between March and August** are because of **COVID-19**. However, it's a very good sign that the **sales increased quickly after August** and reached the highest level since the last two years in November.




**Altogether we have a total sales of 303.92M**

**In FY 2020, we have a total sales of 79.50M**

**In FY 2021, we have a total sales of 224.42M**

Created by Bhavana T

# REQUEST 8:

In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity:

Quarter
total_sold_quantity

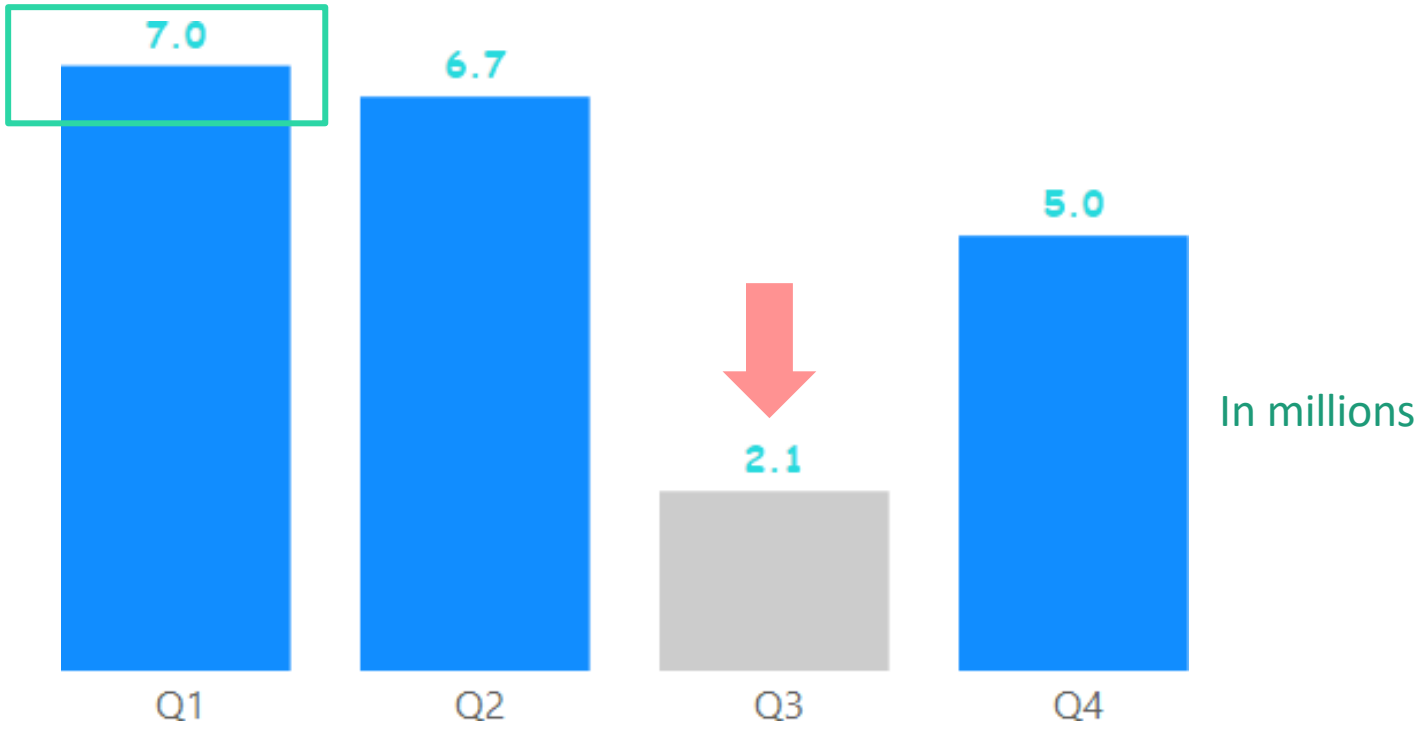
# OUTPUT

| quarter | total_sold_quanity_in_millions |
|---|---|
| Q1 | 7.01 |
| Q2 | 6.65 |
| Q4 | 5.04 |
| Q3 | 2.08 |

# Insights

**How many quantities were sold during the quarters of FY 2020?**

**Quarter 1** has the **maximum** total sold quantity




This again complements the previous insight. That is the effect of COVID-19 on our sales. **The sold quantity decreased to 2.1 million in quarter 3 of FY 2020, which was actually March, April, and May when COVID-19 was at its peak**.

But we started recovering very early despite the continuance of the pandemic. This **early recovery during quarter 4 is probably because of the increased need for hardware like desktops and notebooks as majority of the students began or continued to do their coursework online during this time**, and there was a huge demand for computer accessories during this period.

# REQUEST 9:

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields:

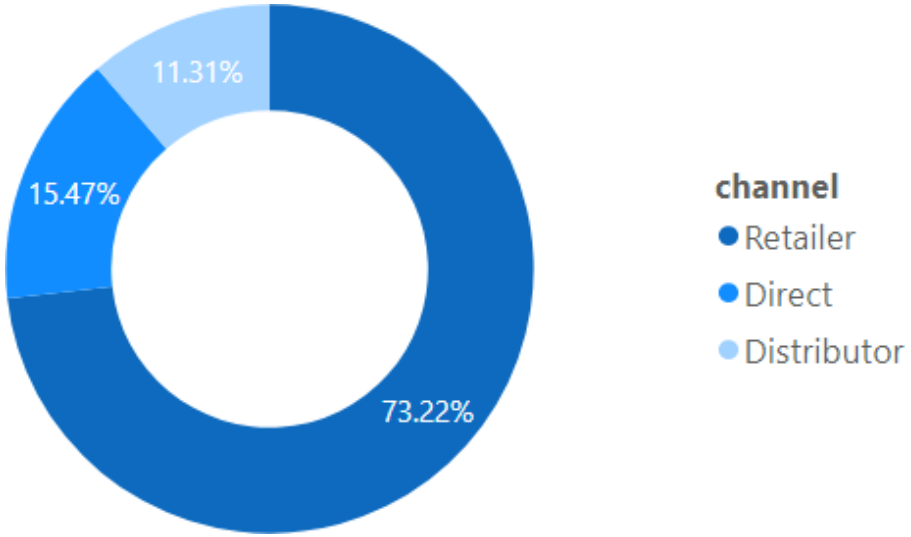
channel
gross_sales_mln
percentage

# OUTPUT

| channel | gross_sales_in_millions | percentage |
|---|---|---|
| Retailer | 1924.17 | 73.22 |
| Direct | 406.69 | 15.47 |
| Distributor | 297.18 | 11.31 |

# Insights

**The majority of our sales** took place **via retailers**, which is **75%** of the total sales. Only a very small percentage of our sales happened through direct and distributor channels.




**In FY 2021, How are our sales distributed among the different channels through which we sell our products?**

# REQUEST 10:

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields:

division
product_code
Product
total_sold_quantity
rank_order

# OUTPUT

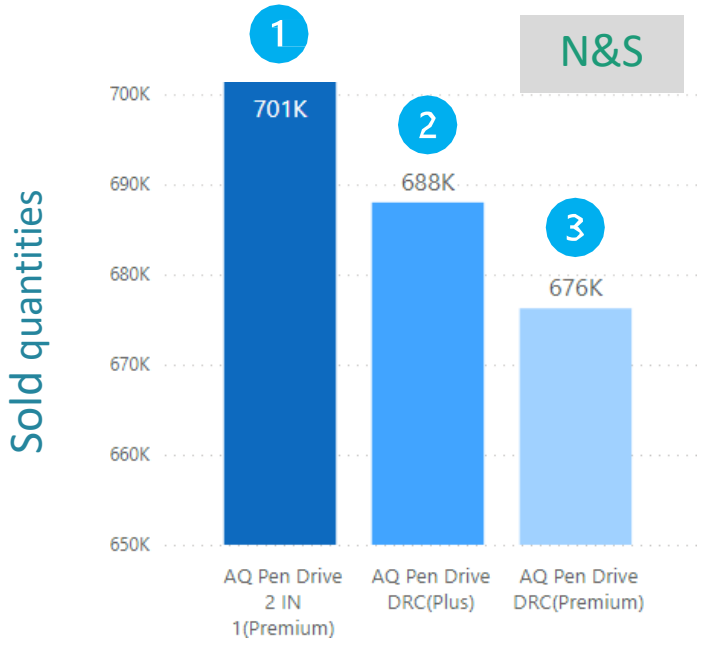
| division | product_code | product | total_sold_quantity | rank_order |
|---|---|---|---|---|
| N & S | A6720160103 | AQ Pen Drive 2 IN 1(Premium) | 701373 | 1 |
| N & S | A6818160202 | AQ Pen Drive DRC(Plus) | 688003 | 2 |
| N & S | A6819160203 | AQ Pen Drive DRC(Premium) | 676245 | 3 |
| P & A | A2319150302 | AQ Gamers Ms(Standard 2) | 428498 | 1 |
| P & A | A2520150501 | AQ Maxima Ms(Standard 1) | 419865 | 2 |
| P & A | A2520150504 | AQ Maxima Ms(Plus 2) | 419471 | 3 |
| PC | A4218110202 | AQ Digit(Standard Blue) | 17434 | 1 |
| PC | A4319110306 | AQ Velocity(Plus Red) | 17280 | 2 |
| PC | A4218110208 | AQ Digit(Premium Misty Green) | 17275 | 3 |

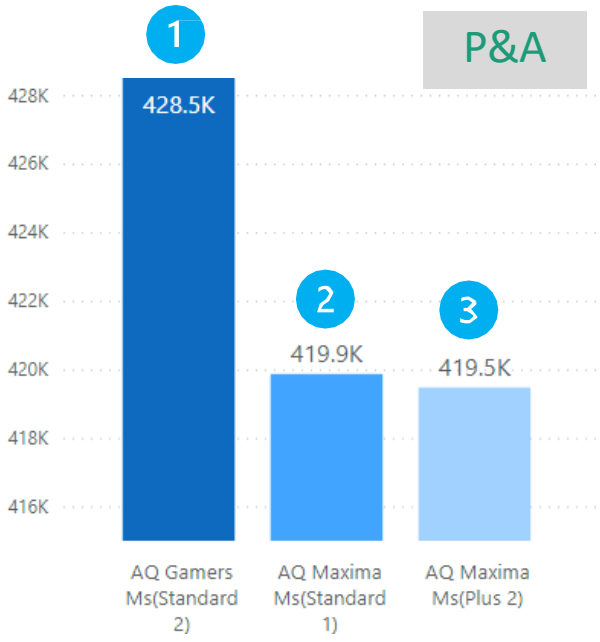
# Insights

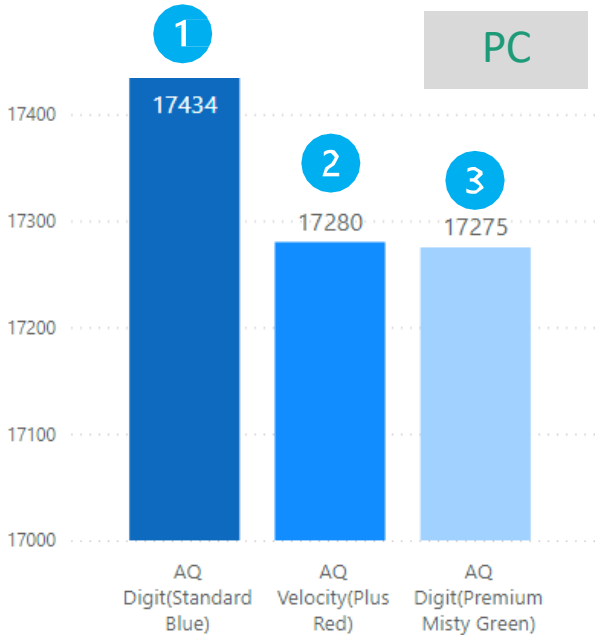
The top 3 selling products in N&S were **pen drives,** which were around **7 lakh in quantity.**

The top 3 selling products in P&A were **mouse,** which were around **4 lakh in quantity.**

The top 3 selling products in PC were **personal laptops,** which were around **17000 in quantity.**




Which are the top 3 selling products in each of Atliq's product lines in FY 2021?

Created by Bhavana T

# END

# THANKYOU!